

Course Objectives

- This course aims to make students, understand, analyse, and appreciate the basic principles, design choices, and trade-offs associated in the field of Computer Architecture.
- It describes overview of ARM architecture in terms of instruction set, data path and pipelining
- It introduces pipelining and memory systems in detail along with performance metrics for designing computer systems and what can be done to make it better and faster.

Course Outcomes

CO1: Understand the design principles of Instruction Set Architecture (ISA) by taking ARM as reference.

CO2: Understand the design of data and control path in Single Clock Cycle.

CO3: Understand the design of instruction execution using Multiple Clock Cycles and Analyze / Evaluate the performance of processors.

CO4: Understand Pipelined architecture and Design of multi-stage pipeline processor in ARM.

CO5: Apply the working principles of ALU, Memory and I/O in the design of processor.

CO-PO Mapping

C O	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
1	2	3	1		3								3	2
2	3	3	3	2									3	2
3	2	2	2		3								3	2
4	2	2	3	2									3	2
5	2	2	2	2	3	2	2						3	2

Syllabus

Unit 1

Overview and history of computer architecture, combinational vs sequential logic, hardware description languages (VHDL), physical constraints (gate delay, fan-in, fan-out, energy/power). Introduction to Instruction Set Architecture, Processor Architecture with example as ARM & Instruction Set, Single Cycle Datapath Design, Control Hardware, Computer Arithmetic, Floating Point Arithmetic, Design of ALU, Introduction and Performance of Computing system and role of performance.

Unit 2

Introduction to multicycle at a path, Instruction level parallelism, instruction pipelining, Pipelining Technique – Design Issues, Hazards: Structural Hazards, Data Hazards and Control Hazards, Static Branch Prediction, Dynamic Branch Prediction, pipeline hazards and advanced concepts in pipelining.

Unit 3

Storage systems, introduction to memory hierarchy: importance of temporal and spatial locality; main memory organization, cache memory: address mapping, block size, replacement, and store policies; virtual memory system: page table and TLB.

External storage; IO fundamentals: handshaking, buffering, programmed IO, interrupt driven IO; Interrupt handling mechanism, Buses: protocols, arbitration. Introduction to modern processors like GPU and TPU, Parallel Processing.

Textbook(s)

J. L. Hennessy and D. A. Patterson (H and P), "Computer Organization and Design, Hardware-Software Interface", ARM Edition , Morgan Kaufmann Publishing Co., 2017.

Reference(s)

L. Hennessy and D. A. Patterson (H and P), "Computer Architecture, A Quantitative Approach", Fifth or Sixth edition, J, Morgan Kaufmann Publishing Co., 2019.

D. A. Patterson and J. L. Hennessy (P and H), "Computer Organization and Design, The Hardware/Software Interface", Fifth edition , Morgan Kaufmann Publishing Co., 2013.

J.P. Hayes, "Computer Architecture and Organization", Mc Graw Hill 2017

William Stallings, "Computer Organization and architecture – Designing for performance", 10th Edition, 2016, Pearson Education.

Evaluation Pattern: 50:50

Assessment	Internal	End Semester
Midterm	20	
*Continuous Assessment (CA)	30	
End Semester		50

*CA includes Quizzes and Tutorials